CYBERSECURITY & NETWORKING FUNDAMENTALS

Section A: Cybersecurity Fundamentals

Q1. Explain cybersecurity and why it is important. Explain CIA Traid with one example each.

Cybersecurity is practice of protecting computers, networks, applications, and data from unauthorized access, attacks, damage, and theft.

Importance:

- Protects personal and organizational data.
- Prevents unauthorized access.
- Protects against viruses and attacks.
- Helps to maintain privacy and business contiunity.

CIA Traid

- Confidentiality: Only authorized people can access the information. Example: a password protects a student's account.
- Integrity: Information remains accurate and unchanged without authorization. Example: a student's marks cannot be changed by an unauthorized person.
- Availability: Systems and information are available when needed. Example: a collage websites remains accessible during admissions.

Q2. Differentiate vulnerability, threat, risk, attack with examples.

Terms	Meaning	Example
Vulnerability	A weakness in a system	Old, unpatched software
Threat	Something that could cause harm	A hacker or malware
Risk	Possibility and impact of harm	Data may be stolen because of weak security
Attack	An actual attempt to harm or access a system	A hacker tries to steal a password

A vulnerability is a weakness, a threat is a possible danger, a risk is the chance of danger, and an attack is an attempt to cause damage.

Q3. Explain any five areas of cybersecurity.

- Network security: Protects computer network from unauthorized access and attacks.

- Application security: Protects software and applications from security weaknesses.
- Cloud security: Protects data and services stored on cloud flatworms.
- Endpoint security: Protects device such as computers, laptops, and mobile phones.
- Information security: Protects information from unauthorized access, modification, or losses.
- Physical security: Protects buildings, computers, and other equipments.

Section B: Networking Fundamentals

Q4. Define a computer network. Explain its uses and describe LAN, MAN, WAN, and PAN.

A computer network is a group of interconnected computers and devices that communicate and share resources.

Uses

- Sharing files and printers.
- Accessing the internet.
- Communicating quickly.
- Sharing applications and resources.

Type	Full form and meaning	Example
LAN	Local Area Network – small area	Collage computer lab
MAN	Metropolitan Area Network – city or urban area	Offices connected across city
WAN	Wide Area Network – large geographical area	The internet
PAN	Personal Area Network – nearby personal devices	Phone connected to wireless earbuds

Q5. Explain the purpose of a hub, switch, router, firewall and access point. Give one difference between a switch and a route.

- Hub: Connects devices and sends incoming data to all its ports.
- Switch: Connects devices in a LAN and forwards data to the appropriate device using MAC addresses.
- Router: Connects different networks and forwards packets using IP addresses.
- Firewall: Monitors and filters network traffic according to security rules.
- Access point: Allows wireless devices to connect to a networking using wi-fi.

Difference: A switch mainly connects devices to connects devices within the same local network, whereas a router connects different networks, such as home network and the internet.

Q6. Explain how the internet works when a user enters example.com in a browser. Include DNS, IP addressing, gateway/router, TCP/IP and web server. Draw a flow diagram.

- DNS: Finds the IP addresses associated with the domain name.
- IP addressing: Identifies the server the browser wants to contact.
- Gateway/Router: Forwards packets from the local network toward the destination.
- TCP/IP: IP routes packets; TCP provides reliable delivery for connections that use TCP, HTTPS commonly uses TCP with TLS, while HTTP/3 uses QUIC over UDP.
- Web server: Receives the request and sends back the webpage.
- Browser: Processes the response and displays the webpage.

1) User enters example.com in browser
2) DNS finds the server IP address
3) Routers/Gateways forward packets
4) TCP/IP or applicable transport protocols carry the request
5) Webserver processes request and returns response
6) Browser displays the webpage

Section C: IP addressing

Q7. Differentiate IPv4 and IPv6. Explain why IPv6 was introduced and give one example of each.

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Format	Decimal numbers	Hexadecimal numbers
Separator	Dots(.)	Colons (:)
Address space	About 4.3 billion addresses	Much larger address space
Example	192. 168. 1. 10	2001:db8::1

IPv6 was introduced because IPv4 addresses were becoming insufficient as internet-connected devices increased. IPv6 provides a much larger address space other improvements.

Q8. IP classification – Identify the IPv4 class and state whether each address is private or public.

IP address	Class	Private/public
10.10.10.5	Class A	Private
172.16.5.20	Class B	Private
192.168.1.50	Class C	Private
8.8.8.8	Class D	Public

IPv4 Classes: Class A = 1-126; Class B = 128-191; Class C = 192-223; Class D = 224-239; Class E = 240-255.

Private ranges are 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

Q9. For 192.168.20.45/24, identify the network address, host portion, subnet mask, broadcast address and usable host range. Show your working.

Given: 192.168.20.45/24

1. Subnet mask: /24 means 24 network bits, so the mask is 255.255.255.0.
2. Network address: Set host bits to 0 ---- 192.168.20.0.
3. Host portion: The last 8 bits (last octet); host value=45.
4. Broadcast address: Set host bits to 1 ----- 192.168.20.255.
5. Usable hosts: 2-2 = 254; Range=192.168.20.1—192.168.20.254.

Item	Answer
Network address	192.168.20.0
Host portion	45
Subnet mask	255.255.255.0
Broadcast address	192.168.20.255
Usable host range	192.168.20.1-192.168.20.254
Total usable hosts	254

Section D: Subnetting & Calculations

Q10. For 192.168.10.0/26, calculate the subnet mask, number of /26 subnets from /24, usable host per subnet, network address, first usable, last usable and broadcast address.

1. Subnet mask: 255.255.255.192.
2. Number of subnets from /24: Borrowed bits = 26-24 =2; subnets = 2^2=4.

3. Usable hosts per subnet: Host bits = 32-26 = 6: Usable hosts = 2^6-2 = 62.
4. Block size: 256-192=64; subnets increase by 64 in the last octet.

Subnet	Network address	First usable	Last usable	Broadcast
1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192	192.168.10.193	192.168.10.254	192.168.10.255

Final answer: Mask = 255.255.255.192; number of subnets =4; usable hosts per subnet =62.

Q11. An organization has 192.168.10.0/24 and requires separate networks for IT (50 hosts), HR (30 hosts), Finance (20 hosts), and management (10 hosts). Design an appropriate subnetting scheme.

Use VLSM (Variable Length Subnet Masking) to allocate different subnet size according to host requirements:

- IT: 50 hosts - /26 provides 62 usable hosts.
- HR: 30 hosts - /27 provides 30 usable hosts.
- Finance: 20 hosts - /27 provides 30 usable hosts.
- Management: 10 hosts - /28 provides 14 usable hosts.

Department	Network/CIDR	Subnet mask	Usable host range	Broadcast	Hosts
IT	192.168.10.0/26	255.255.255.192	192.168.10.1-.62	192.168.10.63	62
HR	192.168.10.64/27	255.255.255.224	192.168.10.65-.94	192.168.10.95	30
Finance	192.168.10.96/27	255.255.255.224	192.168.10.97-.126	192.168.10.127	30
Management	192.168.10128/28	255.255.255.240	192.168.10.129-.142	192.168.10.143	14

Conclusion: All the four departments have separate, non-overlapping subnets with enough usable IP addresses.

Section E: Scenario-Based Questions

Q12. A company has 45 employees, 5 servers and 10 networks devices. Design a basic network. State network type, required devices, role of routers/switch/firewall, and explain how an employee;s request reaches a website.

1.Network type: LAN (Local area network) connects employees, servers and devices inside the company. A router connects the LAN to the internet.

2. Required devices:

- Router for internet connectivity.
- One or more switches for computers and servers.
- Firewall to protect the network.
- Wireless access points if wi-fi is needed.
- 5 servers, 45employee computers and 10 other network devices.
- Ethernet cable and network interface cards where required.

3. IP addressing: There are 60 endpoint addresses (45+5+10). A/26 subnet provides 62 usable IPv4 addresses, enough for 60 devices plus one gateway address, leaving one usable address spare. Example: 192.168.10.0/26; subnet mask: 255.255.255.192; example gateway: 192.168.10.1.

4. Network diagram:

Internet/website
Router + Firewall
Network Switch
Employee PCs (45) | servers (5) | Other devices (10)

5.Device roles:

- Router: Forwards packets between the company LAN and external networks.
- Switch: Connects local computers, servers and other devices.
- Fire wall: Filters traffic according to security rules and blocks unauthorized connections.

6.How an employee accesses a website:

- The employee enters a website address in the browser.
- DNS resolves the domain name to an IP address.
- The computers sends the request through the switch.
- The router forwards it through the firewall to the internet.

- The web server processes the request and returns a response.
- The browser displays the webpage.

Q13. A user has 192.168.1.25/24 and can communicate with another local computers but cannot access an external website. List troubleshooting checks in order, including IP configuration, gateway, DNS, connectivity testing and router role.

The problem may be caused by incorrect network settings, gateway failure, DNS problem or router/Internet connectivity.

1.Check IP configuration: Verify IP address, subnet mask and default gateway. On windows run: ipconfig/all

2.Test local connectivity: Ping another local computer or the gateway. Example: ping 192.168.1.1. If the gateway does not respond, check cable/wi-fi, switch, gateway address and firewall. A failed ping alone does not prove the gateway is down because ICMP may be blocked.

3.Test internet connect using IP address: Try: ping 8.8.8.8. If the gateway responds but internet connectivity fails, investigate the router WAN connection, ISP connection, routing and firewall rules. ICMP may be blocked, so a failed ping is not conclusive.

Ping 8.8.8.8

Check DNS: Run: nslookup example.com. If IP connectivity works but resolution fails, check the configured DNS server and DNS settings.

nslookup example.com

5.Test the website: If DNS works but website is still inaccessible, Check HTTPS/HTTP connectivity, browser proxy settings, firewall rules website availability.

6.Check the router: Ensure it has working WAN connection, a valid route to the internet, correct NAT configuration where required, and no rules blocking traffic.

Conclusion: Troubleshoot in this order: IP configuration – local connectivity – default gateway – external IP connectivity – DNS – website and firewall checks – router and ISP.

THE END